ECA18 – OPTOELECTRONICS AND
OPTICAL COMMUNICATION

CO1 ENGINEERING DESIGN ASSIGNMENT

Standards-Compliant Optical Fiber Link Design:
From Communication Requirement to Deployment Decision

Submitted by

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Course Outcome: CO1 – Apply guided-mode, loss, dispersion, compensation and OSNR concepts to optical-fiber link design.

Academic Engineering Design Report

1. Introduction and Design Requirement

The proposed communication link must carry 10 Gb/s traffic over 60 km at 1550 nm between a central office and a business building. The design must maintain adequate received power, pulse integrity, polarization performance and optical signal quality. It should also provide a practical path towards a future 100 Gb/s upgrade.

Design requirement statement: Design a 60 km, 1550 nm single-mode optical link that supports 10 Gb/s with at least 6 dB receiver-system margin, controls entry-bend loss, meets dispersion and PMD limits, maintains OSNR of at least 17 dB, and uses standard ITU-T-compatible fiber and components.

1.1 Important optical concepts

• At 1550 nm, silica fiber has relatively low attenuation and is compatible with established long-haul optical components and amplifiers.

• Ray theory is useful for visualizing acceptance and propagation in large-core multimode fiber. Wave optics and LP-mode analysis are necessary when the core is small and only a few modes can propagate.

• The electric field is guided by total internal reflection; refractive-index profile determines the allowed modes. Coherence and polarization affect interference, polarization-mode dispersion and coherent detection.

• The link must control absorption, Rayleigh scattering, macrobending/microbending, connector and splice losses, chromatic dispersion, PMD and ASE noise.

• A practical design must consider bend radius, installation workmanship, optical power margin, monitoring and upgrade compatibility.

2. Stage 2 – Fiber and Propagation Configuration

2.1 Numerical aperture

The numerical aperture (NA) measures the light-acceptance capability of a fiber in air. For a step-index fiber:

$$NA = \sqrt{(n_{\blacksquare}^2 - n_{\blacksquare}^2)}$$

Fiber A: $NA = \sqrt{(1.4500^2 - 1.4450^2)} = \mathbf{0.1203}$.

Fiber B: $NA = \sqrt{(1.480^2 - 1.465^2)} = \mathbf{0.2102}$.

Fiber B has the larger NA and accepts a wider range of launch angles. A larger NA does not by itself make a fiber suitable for a 60 km, high-bit-rate link; modal dispersion must also be considered.

2.2 V-number and guided modes

$$V = (2\pi a/\lambda)\, NA;\ \text{at } \lambda = 1.55\ \mu m.$$

Fiber A: V = 1.951. Fiber B: V = 21.300.

For a step-index fiber, the single-mode cutoff condition is $V < 2.405$. Fiber A has V = 1.951, so it is single-mode at 1550 nm. Fiber B has V = 21.300, so it is multimode.

Fiber A cutoff wavelength: $\lambda c = 2\pi aNA/2.405 = 1.257\ \mu m$. Since 1.55 µm is greater than λc, only the fundamental mode is guided.

Approximate mode count for Fiber B: step-index $M \approx V^2/2 = 227$ modes; graded-index $M \approx V^2/4 = 113$ modes. These are high-V approximations.

2.3 Fiber comparison and ITU-T selection

Fiber option	Propagation / dispersion	Application suitability
Single-mode step-index (Fiber A)	Only fundamental mode; negligible intermodal dispersion; high bandwidth-distance capability.	Suitable for long-distance high-speed transmission.
Step-index multimode (Fiber B)	Many modes; large differential modal delay and pulse spreading.	Not suitable for 60 km at 10 Gb/s.

Fiber option	Propagation / dispersion	Application suitability
Graded-index multimode	Index profile reduces modal delay by making outer paths faster.	Better than step-index multimode, but not the preferred 60 km solution.
G.652.D	Standard single-mode fiber; dispersion about 17 ps/(nm·km) at 1550 nm.	Widely deployed, but larger chromatic dispersion and higher stated entry-bend loss.
G.655	Non-zero dispersion-shifted single-mode fiber; stated D = 4.5 ps/(nm·km).	Lower chromatic dispersion for the given 1550 nm link.
G.657.A	Bend-insensitive single-mode fiber; stated entry bend loss is only 0.1 dB.	Suitable for tight building entry; compatible variants can be used with G.652.D networks.

Design Decision 1: Use single-mode propagation. Select G.655 as the 60 km transmission fiber to reduce chromatic dispersion, and use a short G.657.A bend-insensitive entry section/pigtail at the building entrance. Fiber B is rejected because its large V-number and many modes produce excessive modal dispersion.

Wave-optical LP-mode analysis is the appropriate model for the selected single-mode link. Ray optics is useful as an intuitive model for acceptance and multimode propagation, but it cannot adequately describe the single-mode cutoff and field distribution.

3. Stage 3 – Optical Power Budget

3.1 Fiber attenuation

$$\alpha_{total} = \alpha_{absorption} + \alpha_{Rayleigh} + \alpha_{bending}$$

$\alpha_{total} = 0.04 + 0.11 + 0.03 =$ **0.18 dB/km**.

60 km fiber loss = 0.18 × 60 = **10.80 dB**.

The largest specified distributed fiber-loss mechanism is Rayleigh scattering (0.11 dB/km), followed by absorption (0.04 dB/km) and distributed bending (0.03 dB/km). Therefore, Rayleigh scattering is the dominant specified distributed mechanism.

3.2 Practical link budget

Loss item	Calculation	Loss
60 km fiber	0.18 × 60	10.80 dB
Four connectors	4 × 0.4	1.60 dB
Ten splices	10 × 0.08	0.80 dB
Building bend – G.652.D	Given	0.80 dB
Building bend – G.657.A	Given	0.10 dB

G.652.D case: Total loss = 14.00 dB; received power = +3 – 14.00 = **-11.00 dBm**; sensitivity margin = 9.00 dB. Margin beyond the required 6 dB = 3.00 dB.

G.657.A entry case: Total loss = 13.30 dB; received power = +3 – 13.30 = **-10.30 dBm**; sensitivity margin = 9.70 dB. Margin beyond the required 6 dB = 3.70 dB.

With G.657.A entry bend loss, the practical link loss is 13.30 dB, received power is –10.30 dBm and sensitivity margin is 9.70 dB. This satisfies the 6 dB requirement with 3.70 dB additional margin. The G.652.D bend case also meets the stated 6 dB requirement, but has less margin.

Design Decision 2: Adopt the G.657.A bend-insensitive entry section. The lower 0.1 dB bend loss improves received power by 0.7 dB compared with the stated G.652.D entry case. Optimize fiber routing and bend radius, use clean low-loss connectors, and use quality fusion splices with acceptance testing.

4. Stage 4 – Dispersion and Polarization Management

4.1 Chromatic-dispersion broadening

$\Delta\tau = D \times \Delta\lambda \times L$, where $\Delta\lambda = 0.4$ nm and $L = 60$ km.

Fiber	Broadening $\Delta\tau$	RSS pulse width $\sqrt{(30^2 + \Delta\tau^2)}$	Comparison with 100 ps bit period
G.652.D	408 ps	409.1 ps	Much greater; severe pulse overlap
G.655	108 ps	112.1 ps	Slightly greater; compensation / system-level validation needed

G.652.D produces 408 ps broadening, giving an RSS pulse width of about 409.1 ps, far above the 100 ps bit period. G.655 produces 108 ps broadening and an RSS width of about 112.1 ps. G.655 is substantially better, but the calculated result still exceeds one bit period; dispersion management is therefore required for the specified pulse and spectral width.

4.2 Dispersion-compensating fiber

For first-order compensation, the accumulated dispersion must sum to zero: DmainLmain + DDCF LDCF = 0.

For the selected G.655 link: LDCF = (4.5 × 60) / 100 = **2.70 km**. Added DCF loss = 2.70 × 0.5 = **1.35 dB**.

With the 2.70 km DCF, total loss = 14.65 dB; received power = **-11.65 dBm**; margin over sensitivity = **8.35 dB**, which is 2.35 dB above the required 6 dB.

For comparison, a G.652.D link would require 10.2 km of the stated DCF and add 5.1 dB loss. Its received power with the G.657.A entry bend value would be –15.40 dBm and its margin would be 4.60 dB, below the required 6 dB. This is another reason to prefer G.655 for the given assumptions.

4.3 PMD and birefringence

Link PMD = PMD coefficient × $\sqrt{L}$ = 0.1 × $\sqrt{60}$ = **0.775 ps**.

At 10 Gb/s, Tb = 100 ps and the limit is 0.1Tb = 10 ps. At 100 Gb/s, Tb ≈ 10 ps and the limit is 1 ps. The calculated 0.775 ps is below both limits, but the 100 Gb/s upgrade has only about 0.225 ps of PMD allowance remaining. Low-PMD components, careful cable installation and end-to-end PMD testing are recommended.

Birefringence is the difference in effective refractive index for two orthogonal polarization modes. Beat length is LB = λ/B. At λ = 1.55 μm:

Standard fiber: LB = 1.55 × 10■■ / 10■■ = **15.5 m**. PM fiber: LB = 1.55 × 10■■ / (4 × 10■■) = **3.875 mm**.

Polarization-maintaining (PM) fiber has deliberately high birefringence and preserves a launched linear polarization when aligned to its principal axis. It is normally needed for polarization-sensitive components or interferometric systems, not throughout an ordinary intensity-modulated telecom link. Standard low-PMD single-mode fiber is appropriate here.

Design Decision 3: Use G.655 transmission fiber with a 2.70 km equivalent DCF section for first-order chromatic-dispersion compensation, G.657.A at the tight-bend entry, and low-PMD components. Validate residual dispersion and PMD across the complete 10 Gb/s link and repeat validation for the 100 Gb/s upgrade.

5. Stage 5 – OSNR and Amplification Strategy

Optical signal-to-noise ratio (OSNR) compares optical signal power with amplified-spontaneous-emission (ASE) noise power in a specified reference bandwidth. The assignment specifies a 0.1 nm reference bandwidth.

OSNR(dB) = Psignal(dBm) – PASE(dBm)

Initial OSNR = –16 – (–39) = **23 dB**.

After the amplifier, signal power = –16 + 14 = –2 dBm. ASE noise power = –39 + 16 = –23 dBm.

OSNR after amplifier = –2 – (–23) = **21 dB**.

The amplifier reduces OSNR by 2 dB because the ASE noise increase (16 dB) is 2 dB greater than the signal gain (14 dB). Nevertheless, 21 dB is above the required 17 dB, leaving 4 dB OSNR margin.

Insufficient OSNR lowers receiver Q-factor, increases BER, reduces the usable system margin and can cause errors or loss of service. Practical measures include using a low-noise optical amplifier operated at an appropriate gain,

minimizing ASE accumulation by avoiding unnecessary amplifier stages, reducing passive loss, and monitoring OSNR at the receiver.

Design Decision 4: Accept the stated amplifier setting from the OSNR perspective because the calculated post-amplifier OSNR is 21 dB, exceeding the 17 dB requirement by 4 dB. Confirm gain, output power, saturation and OSNR under the actual channel loading before deployment.

6. Stage 6 – Complete Link Verification

Requirement	Calculated / selected result	Satisfactory?
Modal condition	Single-mode G.655 transmission fiber; G.657.A entry section	Yes
Fiber / passive link loss	13.30 dB before DCF; 14.65 dB including DCF	Yes, within available budget
Received power and margin	–11.65 dBm; margin = 8.35 dB including DCF	Yes; exceeds 6 dB by 2.35 dB
Chromatic dispersion	G.655 broadening 108 ps; DCF length 2.70 km for first-order compensation	Conditionally; verify residual dispersion in system test
PMD	0.775 ps; limits 10 ps at 10 Gb/s and 1 ps at 100 Gb/s	Yes, with limited 100 Gb/s headroom
OSNR	21 dB after amplifier vs 17 dB required	Yes; 4 dB margin

The link meets the stated power and OSNR requirements using the selected configuration. The dispersion calculation indicates that first-order compensation is necessary. The PMD value meets the stated limits, although the 100 Gb/s upgrade is more demanding. Final acceptance must include measurement of insertion loss, end-to-end chromatic dispersion, residual dispersion, PMD, OSNR and BER.

7. Final Integrated System Design

7.1 Adopted design parameters

Parameter	Adopted value / decision
Operating wavelength / distance	1550 nm / 60 km
Line rate	10 Gb/s initial; engineered for future 100 Gb/s evaluation
Main transmission fiber	ITU-T G.655 single-mode fiber
Building-entry fiber	G.657.A bend-insensitive pigtail / entry section
Total distributed attenuation	0.18 dB/km (assignment-given components)
Passive link loss before DCF	13.30 dB
DCF length	2.70 km equivalent, D = –100 ps/(nm·km)
DCF attenuation	1.35 dB
Total loss including DCF	14.65 dB
Received power estimate	–11.65 dBm
Receiver sensitivity / margin	–20 dBm / 8.35 dB
Amplifier	14 dB signal gain; ASE increase 16 dB
Post-amplifier OSNR	21 dB in 0.1 nm reference bandwidth

7.2 Proposed link block diagram

Optical amplifier only if required by measured power / OSNR budget

G.657.A bend-insensitive entry pigtail • fusion splices • controlled launch power

7.3 Deployment and commissioning checks

• Measure transmitter output power, connector insertion loss, splice loss and total end-to-end insertion loss.

• Inspect and clean connectors; enforce the manufacturer's minimum bend radius, especially at the building entry.

• Measure chromatic dispersion and confirm that the installed compensating section produces acceptable residual dispersion over the operating wavelength range.

• Measure PMD and OSNR at the receiver; verify BER at the target line rate and under expected channel loading.

• For the 100 Gb/s upgrade, re-evaluate modulation format, spectral width, receiver tolerance, amplifier noise, nonlinear effects, dispersion slope and PMD. The 10 Gb/s calculation alone does not certify 100 Gb/s operation.

8. Final Engineering Decision

DEPLOYMENT DECISION: YES, subject to commissioning verification and upgrade qualification.

The proposed design uses single-mode propagation, avoiding the severe modal dispersion associated with the multimode candidate. G.655 is selected because its stated chromatic-dispersion coefficient is lower than G.652.D. A G.657.A entry section reduces the specified building-bend loss from 0.8 dB to 0.1 dB. The calculated passive loss including the 2.70 km DCF is 14.65 dB, giving an estimated received power of –11.65 dBm and an 8.35 dB sensitivity margin, which is above the required 6 dB.

The G.655 dispersion-compensation length is 2.70 km under the assignment's first-order model. The link PMD estimate is 0.775 ps, below the stated 10 Gb/s and 100 Gb/s limits. The amplifier produces a post-amplifier OSNR of 21 dB, which exceeds the 17 dB requirement. These results support deployment for the stated 10 Gb/s service, provided field measurements confirm the assumed losses and dispersion.

The 100 Gb/s upgrade should be treated as a planned engineering qualification rather than an automatic guarantee. The PMD margin is comparatively small, and the transmitter, receiver, amplifier, dispersion compensation, nonlinear effects and component bandwidth must be checked against the selected 100 Gb/s technology.

Final recommendation: Deploy a 60 km G.655 single-mode link with G.657.A bend-insensitive building-entry fiber, a 2.70 km equivalent DCF section, low-PMD components and an amplifier strategy that maintains at least 17 dB OSNR. Accept the installation only after optical power, residual dispersion, PMD, OSNR and BER measurements meet the design limits.

9. Conclusion

The engineering analysis identifies single-mode G.655 as the most suitable main transmission fiber for the specified 60 km, 1550 nm, 10 Gb/s link. G.657.A is used at the building entry to tolerate tight bends. The link budget including the DCF gives 8.35 dB receiver sensitivity margin, while the post-amplifier OSNR is 21 dB. A 2.70 km equivalent DCF section compensates the stated G.655 accumulated chromatic dispersion in the first-order model. The PMD estimate is within the given limits. Therefore, the proposed link is technically suitable for the stated 10 Gb/s service, with deployment conditional on field verification and a separate qualification process for 100 Gb/s operation.

10. Calculation Summary

Quantity	Result
NA – Fiber A / Fiber B	0.1203 / 0.2102

Quantity	Result
V-number – Fiber A / Fiber B	1.951 / 21.300
Fiber B approximate step-index mode count	227
Fiber A cutoff wavelength	1.257 μm
Total fiber attenuation / 60 km loss	0.18 dB/km / 10.80 dB
Passive loss with G.657.A entry	13.30 dB
DCF length / added loss	2.70 km / 1.35 dB
Total loss including DCF	14.65 dB
Received power / sensitivity margin	–11.65 dBm / 8.35 dB
G.655 chromatic broadening / RSS width	108 ps / 112.1 ps
PMD estimate	0.775 ps
Initial / post-amplifier OSNR	23 dB / 21 dB

Reference: ECA18 – Optoelectronics and Optical Communication, CO1 Engineering Design Assignment (provided question paper). Calculations use the numerical values and simplified models specified in the assignment.